

Self-test Exercise

1. A stochastic process refers to a:
 - a. sequence of random variables indexed by time.
 - b. sequence of variables that can take fixed qualitative values.
 - c. sequence of random variables that can take binary values only.
 - d. sequence of random variables estimated at the same point of time.

2. The sample size for a time series data set is the number of:
 - a. variables being measured.
 - b. time periods over which we observe the variables of interest less the number of variables being measured.
 - c. time periods over which we observe the variables of interest plus the number of variables being measured.
 - d. time periods over which we observe the variables of interest.

3. A static model is postulated when:
 - a. a change in the independent variable at time 't' is believed to have an effect on the dependent variable at period 't + 1'.
 - b. a change in the independent variable at time 't' is believed to have an effect on the dependent variable for all successive time periods.
 - c. a change in the independent variable at time 't' does not have any effect on the dependent variable.
 - d. a change in the independent variable at time 't' is believed to have an immediate effect on the dependent variable.

4. Which of the following is an assumption on which time series regression is based?
 - a. A time series process follows a model that is nonlinear in parameters.
 - b. In a time series process, no independent variable is a perfect linear combination of the others.
 - c. In a time series process, at least one independent variable is a constant.
 - d. For each time period, the expected value of the error u_t , given the explanatory variables for all time periods, is positive.

5. If a time series explanatory variable is strictly exogenous it implies that:
 - a. changes in the lag of the variable does not affect future values of the dependent variable.
 - b. the variable is correlated with the error term in all future time periods.
 - c. the variable cannot react to what has happened to the dependent variable in the past.
 - d. the conditional mean of the error term given the variable is zero.

6. If $\beta_1 > 0$, then y_t in the linear function of time $y_t = \beta_0 + \beta_1 t + u_t$, where the error term is a white noise, displays a(n):

- a. upward trend.
- b. downward trend.
- c. exponential trend.
- d. quadratic trend.

7. A smaller standard error means:

- a. a larger t statistic.
- b. a smaller t statistic.
- c. a larger F statistic.
- d. a smaller F statistic.

8. The Breusch-Godfrey test statistic follows a:

- a. chi-squared distribution.
- b. t distribution.
- c. normal distribution.
- d. F distribution

9. Which of the following is an example of FGLS estimation?

- a. Dickey-Fuller estimation
- b. Maximum Likelihood estimation
- c. Prais-Winsten estimation
- d. OLS estimation.

10. Which of the following is the reason why standard errors measured by OLS differ from standard errors measured through Prais-Winsten transformation?

- a. OLS standard errors account for serial correlation, whereas Prais-Winsten estimations do not.
- b. Prais-Winsten standard errors account for serial correlation, whereas OLS estimations do not.
- c. Prais-Winsten standard errors account for heteroskedasticity, whereas OLS estimations do not.
- d. OLS standard errors account for heteroskedasticity, whereas Prais-Winsten estimations do not.

11. Which of the following is a limitation of serial correlation-robust standard errors?

- a. The serial correlation-robust standard errors are smaller than OLS standard errors when there is serial correlation.
- b. The serial correlation-robust standard errors can be poorly behaved when there is substantial serial correlation and the sample size is small.
- c. The serial correlation-robust standard errors cannot be calculated for autoregressive processes of an order greater than one.
- d. The serial correlation-robust standard errors cannot be calculated after relaxing the assumption of homoskedasticity.

12. In the presence of heteroskedasticity, the usual OLS estimates of:

- a. standard errors are valid, whereas the t statistics and F statistics are invalid.
- b. t statistics are valid, but the standard errors and F statistics are invalid.
- c. F statistics are valid, but the standard errors and t statistics are invalid.
- d. standard errors, t statistics, and F statistics are invalid.

13. A process is *strict* stationary if:

- a. any collection of random variables in a sequence is taken and shifted ahead by h time periods; the joint probability distribution changes.
- b. any collection of random variables in a sequence is taken and shifted ahead by h time periods, the joint probability distribution remains unchanged.
- c. there is serial correlation between the error terms of successive time periods and the explanatory variables and the error terms have positive covariance.
- d. there is no serial correlation between the error terms of successive time periods and the explanatory variables and the error terms have positive covariance.

14. Covariance stationary sequences where $\text{Corr}(x_t, x_{t+h}) \rightarrow 0$ as $h \rightarrow \infty$ are said to be:

- a. unit root processes.
- b. trend-stationary processes.
- c. serially uncorrelated.
- d. weakly dependent.

15. A covariance stationary time series is weakly dependent if:

- a. the correlation between the independent variable at time 't' and the dependent variable at time 't + h' goes to ∞ as $h \rightarrow 0$.
- b. the correlation between the independent variable at time 't' and the dependent variable at time 't + h' goes to 0 as $h \rightarrow \infty$.
- c. the correlation between the independent variable at time 't' and the independent variable at time 't + h' goes to 0 as $h \rightarrow \infty$.
- d. the correlation between the independent variable at time 't' and the independent variable at time 't + h' goes to ∞ as $h \rightarrow \infty$.

16. A stochastic process $\{x_t: t = 1, 2, \dots\}$ with a finite second moment $[E(x_t^2) < \infty]$ is covariance stationary if:

- a. $E(x_t)$ is variable, $\text{Var}(x_t)$ is variable, and for any $t, h \geq 1$, $\text{Cov}(x_t, x_{t+h})$ depends only on 'h' and not on 't'.
- b. $E(x_t)$ is variable, $\text{Var}(x_t)$ is variable, and for any $t, h \geq 1$, $\text{Cov}(x_t, x_{t+h})$ depends only on 't' and not on h.
- c. $E(x_t)$ is constant, $\text{Var}(x_t)$ is constant, and for any $t, h \geq 1$, $\text{Cov}(x_t, x_{t+h})$ depends only on 'h' and not on 't'.
- d. $E(x_t)$ is constant, $\text{Var}(x_t)$ is constant, and for any $t, h \geq 1$, $\text{Cov}(x_t, x_{t+h})$ depends only on 't' and not on 'h'.

17. The model $y_t = u_t + a_1 u_{t-1} + a_2 u_{t-2}$, where $t = 1, 2, 3, \dots$, where u_t is an i.i.d. sequence with zero mean and variance σ^2 represents a(n):

- a. static model.
- b. moving average process of order one.
- c. moving average process of order two.
- d. autoregressive process of order two.

18. The model $x_t = a_0 + a_1 x_{t-1} + u_t$, where u_t is an i.i.d. sequence with zero mean and variance σ^2 represents a(n):

- a. moving average process of order one.
- b. moving average process of order two.
- c. autoregressive process of order one.
- d. autoregressive process of order two.

19. Which of the following statements is true?

- a. A model with a lagged dependent variable cannot satisfy the strict exogeneity assumption.
- b. Stationarity is not critical for time series when performing for OLS.
- c. Static models can be estimated for nonstationary time series.
- d. In an autoregressive model, the dependent variable in the current time period varies with the error term of previous time periods.

20. Economic time series are outcomes of random variables.

- a. True
- b. False

21. In a static model, one or more explanatory variables affect the dependent variable with a lag.

- a. True
- b. False

22. Dummy variables can be used to address the problem of seasonality in regression models.

- a. True
- b. False

23. Covariance stationarity focuses only on the first two moments of a stochastic process.

- a. True
- b. False